



CSC 113 Project

-Phase1-

Report

Members:

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Worked on: class: Member, Volunteer, BoardMember
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Worked on: class: Club, Report
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Worked on: class: Event, Committee

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Project idea:

A club management system.

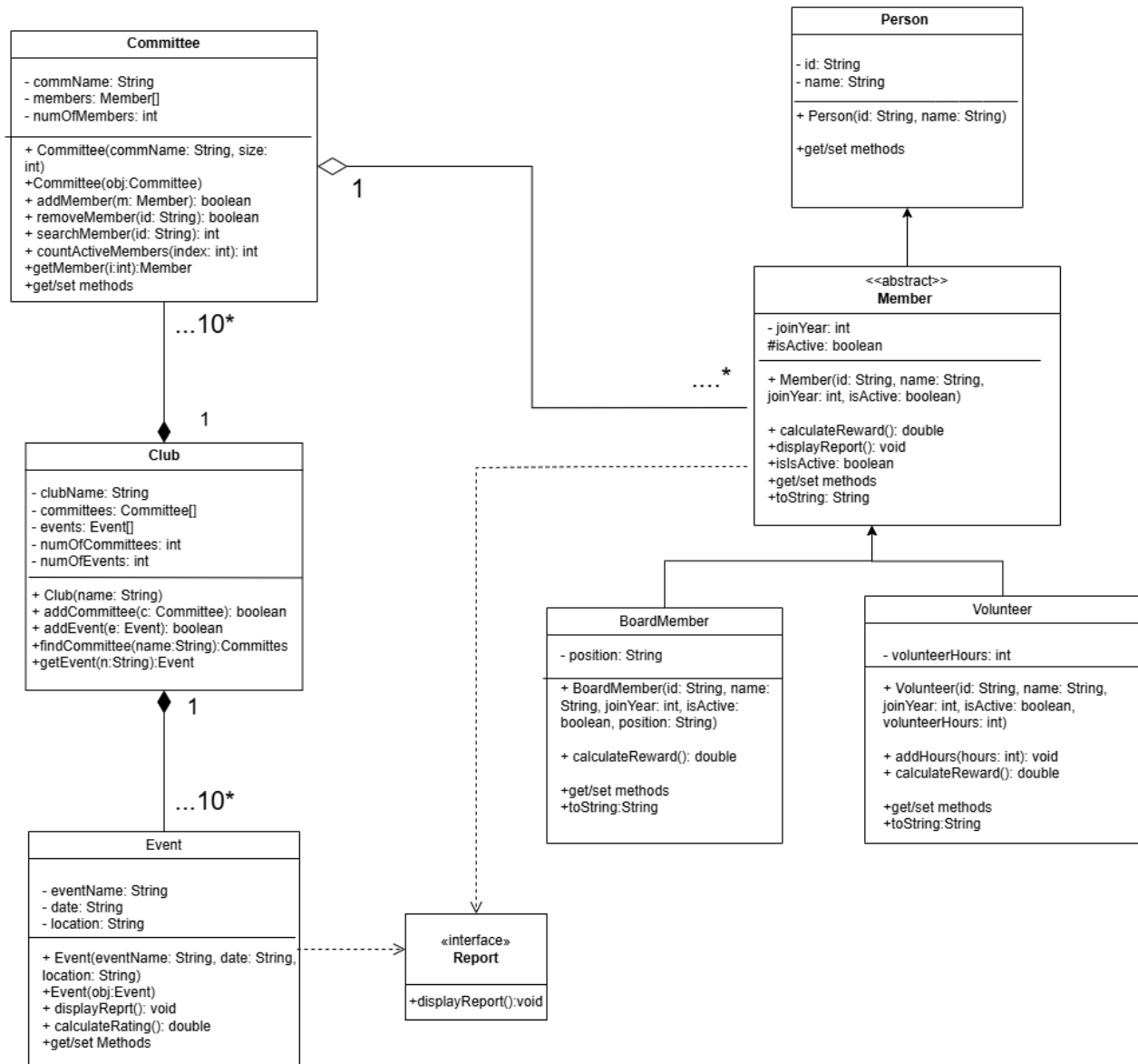
Introduction:

This project is a club management system designed to organize and manage different aspects of a club, including committees, members, and events. The system allows users to add and manage committees, register different types of members such as board members and volunteers, and perform operations like adding or removing members, editing their information, and displaying detailed reports.

In addition, the system provides functionality to calculate rewards for members based on their roles, experience, or participation, as well as count active members using recursion. It also supports event management by allowing users to add events and display their reports along with evaluations.

Overall, the system aims to simplify club operations by providing an organized and efficient way to manage data through a user-friendly menu-driven interface.

UML diagram:



Classes and methods:

Class Club:

Methods:

-public boolean addCommittee (Committee c);

Adds a new committee to the club if there is available space and returns true if added or false otherwise.

-public boolean addEvent (Event e);

Adds a new event to the club if there is available space and returns true if added or false otherwise.

-public Committee findCommittee(String name);

Searches for a committee by name and returns it if found, or null if it does not exist.

-public Event getEvent (String n);

Searches for an event by name and returns it if found, otherwise prints a message and returns null.

Class Board Member

Methods:

-public double calculateReward();

Returns 0 if the member is inactive. If active, it calculates the reward based on two main things: their years of experience, and a specific multiplier for their role (leader, assistant and coordinator).

-public String toString();

Returns a formatted string containing all the specific data of the object...

Class Committee

Methods:

-public boolean addMember(Member m)

Adds a new member if there is space and no duplicate ID exists, returning true if added or false otherwise.

-public boolean removeMember(String id)

Removes a member by ID if found and returns true, otherwise returns false.

-public int searchMember(String id)

Searches for a member by ID and returns its index if found or -1 otherwise.

-public Member getMember(int i)

Returns the member at the given index or null if the index is invalid.

-public int countActiveMembers(int index)

Recursively counts and returns the number of active members in the committee.

Class Event

Methods:

-public double calculateRating()

Calculates the event rating based on its date, returning higher values for recent events or 0 for invalid input.

-public void displayReport()

Displays the event details along with its calculated rating.

Class Member

Methods:

public abstract double calculateReward()

Defines an abstract method to calculate reward, which must be implemented by subclasses.

-public String toString()

Returns a formatted string containing the member's details.

-public void displayReport()

Displays the member's details using the toString method.

Class Person

Methods:

-public String toString():

Returns a formatted string containing all the specific data of the object.

Class Volunteer

Methods:

-public double calculateReward()

If inactive, returns 0. If active, multiplies their total hours by 5 (if they worked less than 25 hours) or by 7 (if they worked 25 hours or more).

-public String toString()

Returns a formatted string representing the volunteer's details.

-public void addHours(int h)

Takes a specific number of hours as an input and adds them to the volunteer's current total volunteerHours.

Sample run:

```
===== the Menu =====
1- Add a new Committee
2- Add a Member
3- Remove a Member
4- Display Member Report
5- Calculate Rewards
6- Edit Member Info
7- Count Active Members (Recursive)
8- Add an Event
9- Display Event Report
10- Exit
=====
Enter your choice:
>> 1
Enter committee name :
>> design
Enter committee size :
>> 10
Committee added successfully.
===== the Menu =====
1- Add a new Committee
2- Add a Member
3- Remove a Member
4- Display Member Report
5- Calculate Rewards
6- Edit Member Info
```


7- Count Active Members (Recursive)

8- Add an Event

9- Display Event Report

10- Exit

=====

Enter your choice:

▶▶

2

Enter committee name to add member to :

▶▶

design

choose member type :

1- Board Member .

2- Volunteer .

▶▶

1

Enter member ID :

▶▶

446

Enter member name :

▶▶

cerine

Enter join year :

▶▶

2025

is active (true/false) :

▶▶

true

Choose Board Member position:

1- Leader

2- Assistant

3- Coordinator

▶▶

Enter your choice (1-3): 2

Member added successfully.

===== the Menu =====

7- Count Active Members (Recursive)

8- Add an Event

9- Display Event Report

10- Exit

=====

Enter your choice:

▶▶

2

Enter committee name to add member to :

▶▶

design

choose member type :

1- Board Member .

2- Volunteer .

▶▶

2

Enter member ID :

▶▶

445

Enter member name :

▶▶

hadil

Enter join year :

▶▶

2024

is active (true/false) :

▶▶

false

Enter volunteer hours :

▶▶

250

Member added successfully.

===== the Menu =====

```
===== the Menu =====  
1- Add a new Committee  
2- Add a Member  
3- Remove a Member  
4- Display Member Report  
5- Calculate Rewards  
6- Edit Member Info  
7- Count Active Members (Recursive)  
8- Add an Event  
9- Display Event Report  
10- Exit
```

```
=====
```

```
Enter your choice:
```

- ▶ 3
Enter committee name to Remove member to :
- ▶ design
Enter member ID to remove :
- ▶ 445
Member removed successfully .

```
===== the Menu =====
```

```
1- Add a new Committee
```

```
===== the Menu =====  
1- Add a new Committee  
2- Add a Member  
3- Remove a Member  
4- Display Member Report  
5- Calculate Rewards  
6- Edit Member Info  
7- Count Active Members (Recursive)  
8- Add an Event  
9- Display Event Report  
10- Exit
```

```
=====
```

```
Enter your choice:
```

```
➤
```

```
4
```

```
Enter committee name of the member :
```

```
➤
```

```
design
```

```
Enter member ID :
```

```
➤
```

```
446
```

```
*BOARD MEMBER*.
```

```
Position: Assistant.
```

```
Name: cerine.
```

```
ID: 446.
```

```
Join Year: 2025.
```

```
is Active?: true
```

```
=====
```

```
===== the Menu =====  
1- Add a new Committee  
2- Add a Member  
3- Remove a Member  
4- Display Member Report  
5- Calculate Rewards  
6- Edit Member Info  
7- Count Active Members (Recursive)  
8- Add an Event  
9- Display Event Report  
10- Exit
```

```
=====
```

```
Enter your choice:
```

▶ 5

```
Enter committee name to calculate rewards:
```

▶ design

```
=== Rewards Report ===
```

```
Member: cerine | Reward: 160.0
```

```
===== the Menu =====
```

```
===== the Menu =====  
1- Add a new Committee  
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4- Display Member Report  
5- Calculate Rewards  
6- Edit Member Info  
7- Count Active Members (Recursive)  
8- Add an Event  
9- Display Event Report  
10- Exit
```

```
=====
```

```
Enter your choice:
```

```
➤ 6  
Enter committee name of the member to edit:  
➤ design  
Enter member ID to edit:  
➤ 446  
1- Edit Name  
2- Edit Active Status  
3- Edit Position  
➤ Enter your choice: 1  
➤ Enter the new name: sham  
Name updated successfully!  
===== the Menu =====
```

```
===== the Menu =====  
1- Add a new Committee  
2- Add a Member  
3- Remove a Member  
4- Display Member Report  
5- Calculate Rewards  
6- Edit Member Info  
7- Count Active Members (Recursive)  
8- Add an Event  
9- Display Event Report  
10- Exit
```

```
=====
```

```
Enter your choice:
```



```
7
```

```
Enter committee name to count active members:
```



```
design
```

```
Total Active Members in 'design' is: 1
```

```
===== the Menu =====
```

```

    Enter your choice:
▶ 8
▶ enter event name: eid
▶ enter event date: 1/10/1447
▶ enter event location: Riyadh
Event added succesfully!
===== the Menu =====
1- Add a new Committee
2- Add a Member
3- Remove a Member
4- Display Member Report
5- Calculate Rewards
6- Edit Member Info
7- Count Active Members (Recursive)
8- Add an Event
9- Display Event Report
10- Exit
=====
    Enter your choice:
▶ 9
▶ enter event name : eid
===== Event Report =====
| name : eid
| date  : 1/10/1447
| location : Riyadh
| evaluation : 3.0 stars
=====

```



```
===== the Menu =====
1- Add a new Committee
2- Add a Member
3- Remove a Member
4- Display Member Report
5- Calculate Rewards
6- Edit Member Info
7- Count Active Members (Recursive)
8- Add an Event
9- Display Event Report
10- Exit
=====
Enter your choice:
10
thank you
```